

Girishkumar Patchikoru

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skills

Programming Languages Python,Java,JavaScript(ES6+),TypeScript, Powershell, React.js, Node.js, Redux, Spring Boot
Cloud & DevOps AWS(lambda,S3,EC2,CloudFormation),Azure,GoogleCloud(GCP), Docker, Kubernetes, Terraform, Jenkins,CI/CD
Databases PostgreSQL, MongoDB, Redis, BigQuery, DynamoDB, Cassandra
Tools & Technologies Git, Jira, Apache Kafka, Spark, Nginx, Webpack
System Design Microservices, Distributed Systems, Load Balancing, Sharding, Event-Driven, Architecture, API Design

Education

Texas A&M University Aug 2023 – May 2025
Master of Science in Computer Science -GPA: (3.8/4.0) *Texas, USA*

Experience

Meta IT Group Aug 2025-present
Mlops Engineer *Newyork,USA*

- Developed, and maintained end-to-end MLOps pipelines covering data ingestion, feature engineering, model training, validation, deployment, and monitoring, automating ML workflows with Python, MLflow, Kubeflow, and Apache Airflow to improve scalability and operational efficiency.
- Built and managed CI/CD/CT pipelines using Jenkins, GitHub Actions, GitLab CI/CD, and Azure DevOps, containerized ML applications with Docker, and deployed scalable inference services on Kubernetes, OpenShift, and managed cloud platforms with blue-green deployments, canary releases, and rollback mechanisms.
- Managed cloud-based AI/ML infrastructure across AWS, Azure, and GCP using SageMaker, Vertex AI, Azure ML, EKS, AKS, and GKE, provisioned environments via Terraform, CloudFormation, and Ansible, and developed batch and real-time data pipelines with Apache Spark, PySpark, Kafka, and Databricks.
- Implemented model versioning, experiment tracking, and registry solutions with MLflow, built monitoring and observability for model drift, data drift, and system health using Prometheus, Grafana, ELK Stack, OpenTelemetry, and Datadog, and secured ML platforms with IAM, RBAC, OAuth2, HashiCorp Vault, and secrets management.

certis Solutions Pvt Ltd Feb 2022–Aug 2023
Devops Engineer *Pune, India*

- Backend Optimization & Automation – Led optimization efforts within the Core Platform DevOps team, reducing backend workload and improving system efficiency by automating Build Deployment failure resolutions.
- Feature Development & Efficiency Boost – Introduced a new feature in the ICM version section of Build Deployment using a String feature, minimizing effort and enhancing workflow efficiency.
- Cross-Team Collaboration & Issue Resolution – Provided critical support to 50+ internal product and delivery teams, swiftly resolving failures and addressing backlog issues to ensure uninterrupted operations.
- Testing, Documentation & Infrastructure Development – Conducted thorough codebase testing, created comprehensive documentation, and played a key role in developing new infrastructures for testing Power BI report deployments.

Data Consultancy Services Aug 2021–Jan 2022
Full-Stack Engineer *Hyderabad, India*

- Legacy Modernization: Led migration of a monolithic telecom application to a Spring Boot microservices architecture, significantly improving API throughput and enabling faster feature delivery.
- Performance Tuning: Upgraded a JavaScript codebase to ES6 standards, reducing runtime memory usage and earning a Spotlight Award for technical excellence.
- Database Optimization: Restructured Oracle DB schemas and queries, decreasing average query latency and supporting substantial user growth without infrastructure scaling.

Texas A&M University, Information Technology(OnCampus) Jun 2024–May 2025
IT Student support worker *Corpus Christi, Texas*

- High-Availability Systems: Deployed a PostgreSQL cluster with read replicas and automated failover, ensuring 99.95% uptime for a student records system serving 15,000+ users.
- Security Compliance: Implemented FERPA-compliant encryption (AWS KMS) and RBAC, reducing breach risks by 40% and passing ISO 2701 audits.
- Cost Optimization: Reduced AWS spend by 30% via rightsizing EC2 instances and leveraging Spot Fleet for batch processing.

Key IT Services Jul 2024–Nov 2024
Frontend Developer Intern Volunteer(remote) *Alberta, Canada*

- Enhanced customer-facing web applications using React/Next.js, implementing AI-based optimization to reduce page load times by 40% and carbon emissions by 20% through resource-efficient data fetching and code splitting.
- Implemented a modular component library with Redux for state management, improving developer productivity by 30% and ensuring uniform design across multiple client projects.

Projects

Distributed E-Commerce Platform | React, Node.js, AWS, Kafka, Kubernetes

- Architected a high-traffic platform with real-time inventory tracking, order processing, and payment workflows, achieving 99.99% uptime via AWS auto-scaling and Kubernetes-managed micro services.
- Reduced API latency by 45% through Redis caching and Kafka-powered event-driven architecture.
- Implemented machine learning-based fraud detection (Python/TensorFlow), decreasing fraudulent transactions by 30%.

Real-Time Geospatial Analytics Engine | Python, Spark, Google Cloud, BigQuery

- Built a distributed data pipeline processing large volumes of geospatial data using Apache Spark, reducing ETL latency by 60% through parallelized coordinate transformations.
- Visualized insights with dynamic Mapbox dashboards, enabling Fortune 500 clients to optimize logistics routes and cut fuel costs by 25%.
- Integrated anomaly detection (PyTorch) to flag irregular patterns with 92% accuracy.

Kernel Craft: OS Scheduling, Memory Management & File Systems

- Implemented system calls specific to each scheduler and core Unix commands enabling efficient task execution and enhancing process management, contributing to a 20% increase in operational efficiency.
- Analyzed and optimized CPU scheduling techniques to minimize Total Response Time (TRT) and Turnaround Time (TAT), achieving a 22% boost in process execution efficiency and reducing system latency.
- Efficiently utilized disk space, improved system performance by 15%, achieved fast process creation as well as file creation by implementing lazy allocation and copy-on-write fork techniques